MILITARY SPECIFICATION

CARTRIDGE, HVAPDS-T, M331A1, FOR 76MM GUNS, T91 AND T124,
WITH TUNGSTEN-CARBIDE CORE,
LOADING, ASSEMBLING, AND PACKING

1. SCOPE

1.1 This specification covers hypervelocity armor-piercing discarding sabot (HVAPDS) tracer loaded cartridge; the shot cores of which are made from tungsten-carbide and the shot carriers (sabot) of which are made of steel and aluminum.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specification, drawing, and publication, and all drawings and specifications referenced thereon, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS
MILITARY
MIL-G-2550 - General Specification For Ammunition Except Small Arms Ammunition.

DRAWINGS
ORDNANCE CORPS
P-82079 - Cartridge, HVAPDS-T, M331A1, for 76MM Guns, T91 and T124. Assembly, Details and Marking Diagram

PUBLICATIONS
ORDNANCE CORPS
STANDARD INSPECTION PROCEDURE
ORD-SIP-M11-110 - Cartridge, HVAPDS-T, M331A1, for 76MM Guns, T91 and T124, Loaded.

(Copies of drawings, specifications, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. - Materials and parts shall comply with all requirements specified on applicable specifications and drawings.

3.2 The cartridge shall comply with all requirements specified on drawings and in applicable specifications and shall have no defects which would make the material hazardous for use.

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3.3 Propellant.

3.3.1 The weight of the propellant shall not differ by more than plus or minus 0.5 ounce avoidupois from that designated by the loading authorization for the particular lot of propellant used.

3.3.2 The contractor shall weigh and checkweigh each charge of propellant. The checkweighing shall be accomplished by an operator working independently of the original weigher, using a different scale from that used to make the first weighing.

3.4 Crimp. - The cartridge case shall separate from the shot at a load between 2000 pounds minimum and 8,500 pounds maximum.

3.5 Proving ground.

3.5.1 Average velocity. - The difference between the velocities of any two cartridges fired in the velocity test shall not exceed 3 percent of the mean velocity of all the cartridges fired. The average velocity shall not differ from the prescribed velocity by more than plus or minus 2 percent after reducing results to standard conditions.

3.5.2 Tracer functioning. - Not more than one tracer shall fail to burn in flight to a target 1000 yards down range.

3.5.3 No cartridge shall fail to discard its sabot during flight.

3.5.4 A lot of cartridges will be considered to have complied with these requirements if samples selected and tested in accordance with 4.5 comply with the acceptance provisions specified therein.

3.6 Workmanship.

3.6.1 All parts shall be free of chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

3.6.2 Care shall be exercised at all times so that the rotating band of each round is not damaged by being nicked or burred.

3.6.3 The painting and marking shall be restored by the assembling contractor on any components furnished to him painted and marked, when such painting and markings have been marred or obliterated.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot. - A lot shall consist of all cartridges assembled by one manufacturer in one unchanged process in accordance with the same drawing, same drawing revision, same specification and same specification revision. Each lot shall contain:

a. Cartridge cases from not more than one lot of resized cartridge cases, or

b. Brass or steel cartridge cases of new manufacture limited to one drawing and drawing revision and same specification date without regard to manufacturer.

c. Loaded primers from not more than two lots. These lots should be consecutive lots if possible (not a requirement).

d. Propellant from one lot.

e. Tracer loaded shot containing shot made by one manufacturer at one plant in accordance with the same drawing, drawing revision and same specification and specification revision, except that cores may be from as many as 10 core lots from as many as three manufacturers.

f. Tracers from not more than two lots.

4.2 Sampling. - Number of samples, acceptance and rejection criteria used for determination of lot acceptance for test as specified in 4.4 shall be in accordance with ORD-SIP-M11-110. Five cartridges shall be selected from each lot for test as specified in 4.5.

4.3 Inspection.

4.3.1 Inspection shall be as specified in Specification MIL-G-2550.

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4.3.2 Gaging. - After the cartridge has been assembled, the contractor shall gage each cartridge in a profile and alignment gage (commonly known as a chamber gage) pertaining to the cartridge under contract. If the cartridge fails to enter the profile and alignment gage freely, it shall be rejected, subject to correction and retest. Drawings of the Government profile and alignment gage will be furnished to the contractor for his information upon application to the inspector.

4.4 Test of crimping case to shot. - Sample cartridges shall be tested in a tensile machine, satisfactory to the inspector, to determine the security of the crimp of the cartridge case to the shot, by applying a load in the axial direction, rate of travel of testing machine head to be .100 to .125 inch per minute inclusive.

4.5 Proving ground test.

4.5.1 Procedure. - Fire five sample cartridges in a gun for which the cartridge is standard. The tracer functioning shall be determined by suitable means and the velocity shall be measured.

4.5.2 Acceptance. - The lot shall be considered acceptable provided that in the phase being tested, (i.e., velocity or tracer functioning) the cartridges fired and complied with the applicable requirements of 3.5.1, 3.5.3, 3.5.3, and 3.5.4. The lot shall be considered eligible for retest under the provisions of 4.5.3 if:

a. The difference between the velocities of any two cartridges fired exceed 3 percent but does not exceed 4 percent of the mean velocity of all cartridges fired in the phase of test, or

b. The average velocity differs from the prescribed velocity by more than plus or minus 2 percent, but not more than plus or minus 3 percent, or

c. Not more than two tracers fail to burn in flight to the target, or

d. Not more than one cartridge fails to discard its sabot during flight.

The lot shall be rejected provided that in the phase being tested, the limits for eligibility for retest are exceeded.

4.5.3 Retest. - A retest as specified in 4.5.2 may be made if requested by the contractor using double the number of samples used in the phase being tested. The lot shall be considered acceptable on retest provided that in the phase being tested, (i.e., velocity tracer functioning) the cartridges fired and complied with the applicable requirements of 3.5.1, 3.5.2, 3.5.3, and 3.5.4.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking. - Packing, labeling, and marking shall be as specified on the applicable drawing.

5.2 Data cards. - Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Cartridge. - A cartridge, as referred to herein, means the assembly of all the components of ammunition into one unit necessary for firing the cartridge from a gun: example - cartridge case, percussion primer, propellant and tracer loaded shot, all properly assembled.

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6.2 When the contractor for the cartridge is also the contractor for one or more of the components thereof, the proving ground tests of the components may be combined with the proving ground tests of the cartridge to save expense upon agreement between the Government and the contractor to that effect. In case the cartridge specification does not cover all of the proving ground tests specified for the component, the additional tests specified in the component specification should be conducted.

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